Lorenzo **Liuzzo**

PHYSICIST · SOFTWARE ENGINEER · AI SYSTEMS

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Physicist and *software engineer* specialized in *AI*. Physics taught me to derive the model before writing the code; lately most of my time goes into learning how much of that writing can be delegated to a model, and where a person still has to decide.

Experience

MyThingsLab — MyThingsLab

PERSONAL PROJECT — July 2026 - Present

- Somewhere to find out how far AI-assisted development can be taken, and where it should stop: **more than 50 small Python tools** in the first month, all on one SDK — **my-things-core** — exposing five contracts (ledger, policy, engine, GitHub, isolation) that every tool imports and no tool reimplements.
- **Control, not autonomy.** Headless workers pick up an issue, do the deterministic pre-work with no model call, ask the LLM once for the step that needs judgement, and close it as a pull request — never a merge, so the last word stays with a person.
- **No model inside the core.** The SDK is dependency-free and shells out to **gh** and **git**, and the LLM sits behind a single **Engine** protocol whose deterministic default lets the whole fleet be exercised and tested without a token spent.

Money Balling AI — MoneyBallingAI

FOUNDER AND TECHNICAL LEAD — September 2025 - Present

- A team of four building one model that predicts the **exact box score** of an NBA game — every player and team line — with calibrated uncertainty. I am responsible for the architecture, from raw play-by-play to a calibrated prediction:
- **Data** — play-by-play parsed into periods, lineup stints and possessions, aggregated into cumulative stat windows and modelled as heterogeneous stint graphs, with stable global identities for players and teams across seasons.
- **Encoder** — a heterogeneous GNN over those graphs with per-entity cross-game and hierarchical season/game/stint memory, pretrained on a box-score delta task for stable player and team strength embeddings.
- **Simulator** — a vectorized Monte Carlo possession engine whose policies are behavior-cloned on real possessions; **calibration** scores per-stat distributions against the actual box score with MAE, CRPS, PIT and coverage, test-first and under CI.

Private Tutoring in Mathematics and Physics — Milan, Italy

IN-HOME TUTORING — 2020 - Present

Continuous support for 20+ students: exam preparation, remedial tutoring and periodic reports to families.

Barista — Milan, Italy

DON SALVATORE - PIZZAIUOLO E OSTE — September - October 2024

Waiter — Milan, Italy

BAR TERRAZZA CLÉR — May - July 2022

Scout Leader — Milan, Italy

REPARTO BREITHORN - GRUPPO MI8 - AGESCI — 2021 - 2022

Educational activities for youth aged 12-16: camps, outings, coordination with educators and families.

Education

University of Milan, Milan-Bicocca and Pavia — Milan/Pavia, Italy

MASTER'S DEGREE IN ARTIFICIAL INTELLIGENCE FOR SCIENCE AND TECHNOLOGY — September 2025 - Present

University of Milan — Milan, Italy

BACHELOR'S DEGREE IN PHYSICS, GRADE: 100/110 — September 2021 - July 2025

Thesis: *ray tracing and Monte Carlo methods for uncertainty quantification in radio-telescope pointing for CMB observations.*

I.I.S. Severi-Correnti — Milan, Italy

SCIENTIFIC DIPLOMA, GRADE: 100/100 — September 2016 - June 2021

EF International Language School — Brighton, England

FULL-IMMERSION INTENSIVE ENGLISH LANGUAGE COURSE — June - September 2019

Schools & Workshops

School on Quantum Simulation
Milan, Italy

DEPARTMENT OF PHYSICS, UNIVERSITY OF MILAN
September 2026

Qiskit Fall Fest
Milan, Italy

IBM QUANTUM · DEPARTMENT OF PHYSICS, UNIVERSITY OF MILAN
October - November 2025

Projects

Gauge-Equivariant Graph Neural Networks
lorenzoliuzzo/EGNN

DESIGNER/DEVELOPER
July - August 2026

Lattice Standard Model field theory simulated with gauge-equivariant GNNs in **PyTorch Geometric**: SU(3)×SU(2)×U(1) gauge groups, Wilson-Dirac fermions and Higgs spontaneous symmetry breaking.

MSc Coursework Projects
lorenzoliuzzo

DESIGNER/DEVELOPER
May - July 2026

- **Supervised Learning on Food Images** — a **CNN** under 10M parameters classifying the 251 classes of the FoodX-251 dataset.
- **Transverse Field Ising Model** — quantum phase transitions explored on **PennyLane**.
- **Unsupervised Learning on Country Data** — clustering and dimensionality reduction on socio-economic country-level data.

Telescope Pointing Simulation
lorenzoliuzzo/thesis-L30

DESIGNER/DEVELOPER
June - July 2025

Julia model of a telescope's reflective system, propagating mirror defects to the pointing direction via *ray tracing* and *Monte Carlo* simulation.

Systems & Numerical Programming
C++ · Nim

DESIGNER/DEVELOPER
2023 - 2024

- **Ray Tracing Engine** — pair project at UNIMI: a physical model turned into structured, tested and documented numerical code, while rapidly learning a new language (**Nim**).
- **Dimensional Analysis Library** — "CTDA: Compile Time Dimensional Analysis", C++ numeric types carrying units of measurement in the type system.

How I Build Software

- **AI across the whole workflow, not only the coding step.** Composable tools cover research, design, implementation, testing and documentation, so the model is used wherever it earns its place rather than at a single step.
- **Deterministic first, models only where judgement is needed.** The LLM sits behind a single **Engine** seam, so a tool's entire skeleton runs and is tested at zero token cost.
- **Test-driven, and always through a pull request.** Tests are written with the code and often before it; **pytest**, coverage and **ruff** run on every push, and work lands by review rather than by pushing to a main branch.
- **Every side effect passes a gate.** A policy engine rules each action allow/ask/deny, tools open pull requests rather than merging, and every *ask* reaches me over Telegram before anything happens.
- **Understand the problem before writing the code.** A habit kept from physics: work out the model first, decide what the code actually has to compute, then write it — typed, documented, and with no abstraction built for a need that does not exist yet.

Skills

Languages	**Italian**, **English C1**
Programming Languages	**Python**, **C++**, **Julia**, C, Nim
AI Engineering	**Claude**, **Claude Code**, Agentic workflows, Prompt & context engineering
ML & Data Science	**PyTorch**, **PyTorch Geometric**, Scikit-learn, Pandas, NumPy, SciPy
Databases & Quantum	Neo4j, Cypher, **PennyLane**, Qiskit
Practices	**Test-driven development**, **Code review**, CI/CD, Static typing
Development Tools	**Git**, **GitHub**, GitHub Actions, pytest, ruff, Jupyter
Typesetting	**LaTeX**, **Typst**, **Markdown**, Jekyll